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Photocycles of Channelrhodopsin-2

Konstantin Nikolic*[1], Nir Grossman[1], Matthew S. Grubb[2], Juan Burrone[2], Chris Toumazou[1] and Patrick Degenaar[1]

[1]Institute of Biomedical Engineering, Imperial College London, London, UK
[2]MRC Centre for Developmental Neurobiology, King's College London, London, UK

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ABSTRACT

Recent developments have used light-activated channels or transporters to modulate neuronal activity. One such genetically-encoded modulator of activity, channelrhodopsin-2 (ChR2), depolarizes neurons in response to blue light. In this work, we first conducted electrophysiological studies of the photokinetics of hippocampal cells expressing ChR2, for various light stimulations. These and other experimental results were then used for systematic investigation of the previously proposed three-state and four-state models of the ChR2 photocycle. We show the limitations of the previously suggested three-state models and identify a four-state model that accurately follows the ChR2 photocurrents. We find that ChR2 currents decay biexponentially, a fact that can be explained by the four-state model. The model is composed of two closed (C1 and C2) and two open (O1 and O2) states, and our simulation results suggest that they might represent the dark-adapted (C1-O1) and light-adapted (C2-O2) branches. The crucial insight provided by the analysis of the new model is that it reveals an adaptation mechanism of the ChR2 molecule. Hence very simple organisms expressing ChR2 can use this form of light adaptation.

INTRODUCTION

ChR2 is a light-gated cation channel derived from the *Chlamydomonas reinhardtii* algae (1). Its expression in mammalian cells results in depolarizing photocurrents when illuminated with blue light (1,2). The rapid onset of the current together with its unique sensitivity has resulted in a number of experiments and applications which exploit these properties. They include the stimulation of neurons in a network and functional mapping of their connectivity (3–6), the generation of behavioral responses in *Caenorhabditis elegans* through the stimulation of sensory neurons (7), and the induction of visually evoked behavior in rodents through ChR2 expression in retinal ganglion cells (3) and more recently in ON bipolar cells (8). ChR2 offers the best promise for experimental neuroscience to remotely and noninvasively stimulate neurons and has direct implications in future therapeutic strategies (2–4,9,10). Although ChR2 is widely used for neuronal stimulation, a number of issues remain unclear regarding the detailed channel activation response and its complete photocycle.

Corresponding author email: k.nikolic@imperial.ac.uk (K. Nikolic)
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ChR2, like other opsin-based proteins, absorbs light through its tight interaction with retinal. The native chromophore in the dark-adapted ChR2 expressing cells is the all-*trans* retinal (11), which undergoes *trans–cis* isomerization after illumination. It has also been shown that other types of retinal (*e.g.* 13-*cis*, 11-*cis*, *etc.*) can also activate the ChR2 ion channel (11), but with reduced efficiency. The first spectroscopic investigations on ChR-type proteins have only been reported very recently (12,13). The UV–visible spectroscopic data revealed the spectral properties and the photocycle of the activation process for Volvox ChR (12). The results point to the existence of two dark, nonconducting states (D1 and D2) with pH-dependent equilibrium, three fast (ns to μs), early intermediates, and two longer (ms) intermediates with extracellular pH-dependent variations.

Previous studies have analyzed the kinetics of ChR2 currents and possible photocycle models have been proposed (1,12–17). There are currently two major models in the literature: a three-state (1,16) and a four-state model (1), as shown in Fig. 1. Both models can *qualitatively* reproduce the ChR2 photocurrents. The first three-state model of the ChR2 photocycle was proposed by Nagel *et al.* (1,14) and showed that the overall kinetics of the currents, as well as the relative amplitudes of the peak-plateau currents, could be qualitatively reproduced. They also reported that the recovery rates under constant illumination (light adapted) and in the dark (dark adapted) need to be different in order to reproduce the characteristic peak-plateau behavior and degraded transient response for subsequent stimulus. In the same paper a four-state model was postulated to account for these observations. A more detailed investigation followed in Hegemann *et al.* (14), where several four-state models were examined for channelrhodopsin-1 (ChR1). Nikolic *et al.* (16) then proposed a modified three-state model which introduced a light-assisted recovery of ChR2.

Based on these descriptions, Ernst *et al.* (12) proposed a hybrid model, which is a combination of the four-state model and the three-state model proposed by Nikolic *et al.* (16). This model effectively postulates a single open state by assuming a fast equilibrium between the two open states, O1 and O2, equivalent to the open states shown in Fig. 1b. The open state relaxes to two intermediate states (D1 and D2) sequentially. The peak-plateau shape of the photocurrent is explained by lower quantum efficiency for the D2 $\rightarrow$ O transition. Finally, Bamann *et al.* (13) examined the spectral characteristics of ChR2 by activating the channel with a short laser pulse

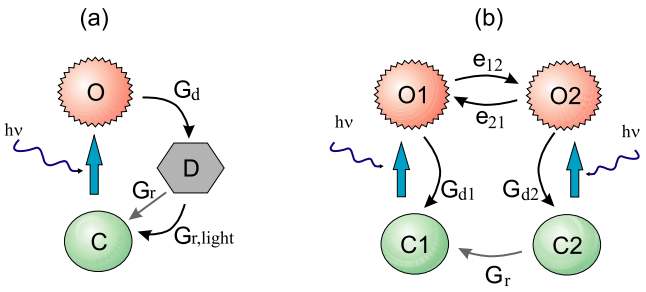


Figure 1. Two simplest models of ChR2 photocycle which can qualitatively reproduce the photocurrents in ChR2 transfected neurons. (a) The three-state model: C—closed/ground, O—open and D—closed/desensitized states. (b) Four-state model, with two closed C1 and C2 and two open states O1 and O2. The transition rate constants are described in the text.

(480 nm, 10 ns) and measuring the absorbance changes in the visible range, as the molecule relaxed back to the dark ground state. The results revealed four different relaxation processes, with very different time constants, leading to the identification of four kinetic intermediates P_1, P_2, P_3 and P_4. P_1 is a short-lived (~0.15 ms) intermediate activated from the dark-adapted P_0 state. P_2 and P_3 are considered to be open states (with time constants of ~1 and ~10 ms, respectively)—possibly we may identify them as O1 and O2 states in the four-state model shown in Fig. 1b. P_4 is a relatively long-lived (~5 s) state which is spectroscopically almost identical to the ground state. This state would probably correspond to the desensitized state in the three-state model (Fig. 1a). The photocycle goes in a circle of ChR2 states which correspond to the following kinetic intermediates: $P_0 \rightarrow P_1 \rightarrow P_2 \leftrightarrow P_3 \rightarrow P_4 \rightarrow P_0$.

In addition to the ChR2 photocycle modeling, a more detailed, quantitative investigation is also needed in order to get a better insight into the photokinetics of ChR2-expressing cells. Here, the light-evoked currents were recorded from hippocampal neurons expressing ChR2. During long light stimuli the current reaches a peak, which decays to a steady-state plateau. During very short light stimuli the current briefly continues to rise after the end of the light pulse, as described previously in Wang *et al.* (6). After the light is turned off, the current decays back to baseline in a biexponential manner. We use the three-state and four-state generic functional models (Fig. 1) to fit our data without any initial assumptions on the lifetime and conductivity of each state. These two models have been described in the literature before, *e.g.* the three-state in Nikolic *et al.* (16) and the four-state in Hegemann *et al.* (14). The results which can be explained by the three-state model (16) are the peak–plateau values and the peak recovery, and we repeat the results for the peak–plateau values here in order to have a complete overview of the achievements of both models. However, Nikolic *et al.* (16) do not analyze the limitations of the model, such as biexponential decay and the need for light-dependent deactivation and recovery rates. In this study we make *quantitative* comparisons of the three-state model and the experimental results for a long pulse, voltage-clump photocurrent, clearly pointing to the shortcomings of the three-state model to provide a good fit. Here we give the results of the analysis of the three-state model which render it unlikely to be an adequate description of the ChR2 photocycle.

Several four-state models are proposed and analyzed by Hegemann *et al.* (14). However, the model was mathematically presented and numerically simulated as a set of difference equations, whereas we use a set of ordinary differential equations (rate equations). The rate equations were exploited not only for numerical simulation but also toward obtaining some analytical results, which are described in Appendix 2 and which help us in choosing the correct parameters. In addition, Hegemann *et al.* (14) is devoted to the analysis of the light-induced conductance of ChR1, not ChR2. Nevertheless from the formal modeling point of view, the four-state model described in Hegemann *et al.* ([14], Fig. 2e) represents a good start for our analysis. Furthermore, the four-state model from Hegemann *et al.* (14) is improved by the introduction of a time-dependent activation rate, described in detail in Appendix 1. This improvement is necessary to explain a ChR2-expressing cell response to short light pulses. Eventually, it is argued that the deactivation of ChR2 has to be light induced in order to explain the experimental fact that the rate of photocurrent decay after the peak is bigger for higher light intensity. As a result we have established the parameters that best fit our experimentally measured currents.

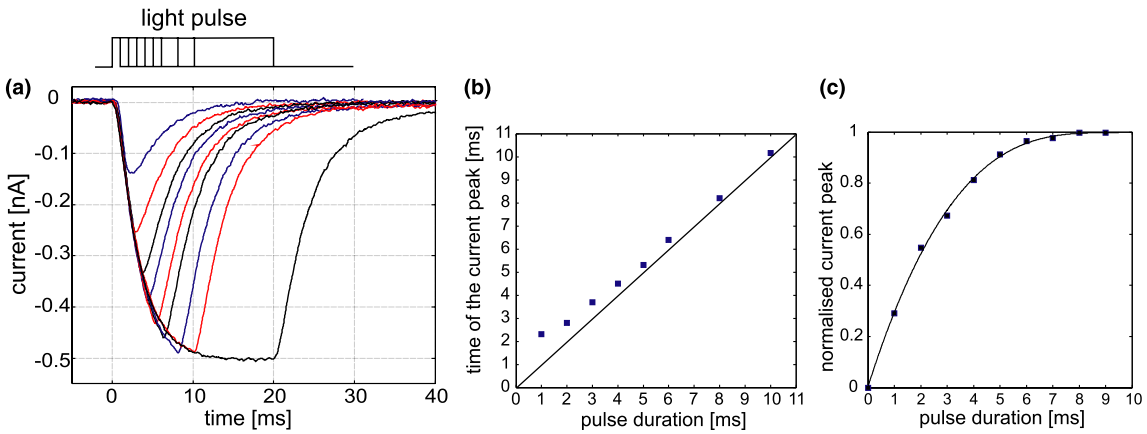


Figure 2. (a) Photocurrents in ChR2-expressing hippocampal cells for short light pulses (1, 2, 3, 4, 5, 6, 8, 10 and 20 ms). Light intensity is approximately 1 mW mm^{-2}. (b) Time to peak photocurrent and (c) normalized values for the peak current *vs* photostimulus duration.

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MATERIALS AND METHODS

ChR2 expression vectors. To insure strong neuronal expression of ChR2, we subcloned the fusion protein ChR2-YFP (a gift from Karl Deisseroth) into a plasmid containing a chick β-actin promoter. For investigation of different fluorophores, mCherry and myc sequences were substituted for the C-terminal YFP sequence.

Cell culture and ChR2 expression. Primary dissociated cultures were obtained from rat hippocampal tissue on embryonic day 18.5. Hippocampi were digested in trypsin (1 mg mL^{-1} in HBSS, 15 min at 37°C), and dissociated in neurobasal medium with 10% FCS by passing through a series of decreasing diameter Pasteur pipettes. They were then plated at $\sim$300 cells mm^{-2} on 18 mm diameter glass coverslips coated in PDL (50 μg mL^{-1}) and laminin (20 μg mL^{-1}). Cultures were grown in neurobasal medium supplemented with 10% FCS, 0.5% penstrep and 1× glutamax for the first 2 h *in vitro*, and thereafter in neurobasal medium with B27, 0.5% penstrep and 1× glutamax, at 37°C and 6% CO_2. Neurons were then transfected with the appropriate ChR2 vector using a lipofectamine procedure at 7 days *in vitro* (DIV), and electrophysiological recordings were performed at $\geqslant$10 DIV. HEK-293 cells were grown in DMEM with 10% FCS and 1% penstrep, split and plated at $\sim$150 cells mm^{-2}. One day later, they were transfected with ChR2-YFP using an overnight lipofectamine incubation, and were recorded at 3 DIV.

Electrophysiology. Individual coverslips were placed in a custom-made recording chamber and bathed with room temperature HEPES-buffered saline (HBS) containing, in mM: 136 NaCl, 2.5 KCl, 10 HEPES, 10 D-glucose, 2 $CaCl_2$, 1.3 $MgCl_2$, 0.001 tetrodotoxin, 0.01 gabazine, 0.01 NBQX and 0.025 APV (285 mOsm, pH 7.4). For HEK-293 cell recordings, the HBS was supplemented with 1 μM all-*trans* retinal. We used an Olympus IX71 inverted microscope equipped with a 40× oil-immersion objective, a xenon light source and appropriate filters for fluorescence imaging, to localize and target fluorescent ChR2-expressing neurons. Patch pipettes, pulled from borosilicate glass (1.5 mm OD, 1 mm ID, 3–4 MΩ) were filled with a solution containing, in mM: 130 K-gluconate or Cs-gluconate, 10 NaCl, 1 EGTA, 0.133 $CaCl_2$, 2 $MgCl_2$, 10 HEPES, 3.5 Na-ATP, 1 Na-GTP (280 mOsm, pH 7.4). Conventional whole-cell patch-clamp recordings were obtained *via* a Heka EPC10 double patch amplifier coupled to Pulse acquisition software. Signals were sampled at intervals of 65–150 μs (6.7–15.4 kHz) and were low-pass filtered using a four-pole Bessel filter at 2.9 kHz (filter 2). The holding potential in voltage-clamp mode was −70 mV, uncorrected for any liquid junction potential (probably in the range of +10 mV) between our internal and external solutions.

Light source and optics. ChR2-expressing cells were photostimulated using two different light sources. Some experiments involved photostimulation *via* a xenon-arc lamp (Lambda LS; Sutter), where light exposure was regulated by a rapid shutter (smartShutter; Sutter) regulated by a Sutter Instruments lambda 10-3 controller, fitted with a 470 ± 20 nm bandpass excitation filter (Chroma Tech. Corp.) and suitable neutral density filters. Other experiments, in particular those requiring short-duration photostimulation, used light stripes consisting of 120 GaN micro-LED stripes operating at 470 nm (18). In the micro-LED array, each stripe is 17 μm wide and 3600 μm long, and the center-to-center spacing is 34 μm. Each LED has a dedicated driver (TB62710F) that is controlled with a microcontroller (PIC18F4550) allowing fast configuration (complete reconfiguration in 20 μs). The constant current value is set using external digital potentiometers on each driver chip, allowing current values from 3 to 90 mA. The LED array is imaged on the cells with a 1:1 4F relay configuration made of two 50 mm triplet lenses (Sill Optics GmbH). The advantage of this optical configuration is the low aberration (peak to valley aberrations of less than 1.5 waves across the entire field) and the 40 mm long working distance that allows good access to the patch-clamping. A TTL output signal was used to synchronize between the recording and the stimulus. The optic power from the micro-LEDs was adjusted with pulse width modulation and/or by changing the injected current and was recorded with a photodiode (OP-2VIS; Coherent) with a laser power meter (FieldMAx2; Coherent). The power of the stimulating light was measured with a photodiode (818-UV; Newport). The power densities were calculated as the power of light divided by the illumination area (0.24 and 0.07 mm^2 in the case of the xenon and the micro-LED, respectively).

Numerical methods. All numerical calculations were performed in Mathworks MATLAB 7.5.0 (R2007b). The experimental data were imported into MATLAB and visually inspected for consistency. Appropriate ranges of data were visually identified for the light-off current and single and biexponential fitting parametric models were applied. For the data fitting the Curve Fitting Toolbox was used. The system of ordinary differential equations was solved using the ODE23 solver. For fitting the model parameters to the data, a simple least-squares routine was applied. First a set of parameters is altered until a visually satisfying fit to the experimental curve is obtained. A limited range of parameter space was then swept to find the best fit, but incrementally increasing parameters and keeping the new set of values if they give better fit than the previous one.

RESULTS

The ChR2 photocurrent

Because the ChR2 photocurrent has already been well characterized (*e.g.* [6,15]), our purpose here is to simply highlight the major components of the photoelectrophysiological response. These components then instruct our quantitative modeling of ChR2 kinetics in the following sections. Under whole-cell voltage clamp recording, the response of a ChR2-expressing neuron to a stationary blue light stimulus has three main features: an initial peak with a fast decay, a steady-state plateau, and a postillumination decay back to baseline. Subsequent stimulus, applied shortly after the previous one was terminated ($<$10 s), induces a degraded transient response and the same steady-state one.

Under continuous illumination, the value of the peak current rises in a nonlinear fashion with light intensity. However, under short pulsed stimulation (1–10 ms) and lower light intensities the current can show a different type of peak, which appears because the number of active ChR2 channels have not reached the maximum value before the light was switched off (see Fig. 2a). In this case both the value of the peak current and the time to peak vary with pulse duration (Fig. 2b,c). Experiments with very short light pulses (1–5 ms) suggest that some ChR2 channels open after the light stimulus has terminated (Fig. 2b). For example, in the case of a 1 ms light pulse, the current peak occurred approximately 1.5 ms after the end of the stimulus. This is in agreement with spectroscopic measurements (12,13), which show that the activation of ChR2 goes through several intermediates and takes about 0.2–1 ms.

Useful information comes from the final part of the ChR2 photocurrent curve, when the light is turned off (see Fig. 3a top left inset). At the termination of the light stimulus, after an arbitrary long light stimulation, the ChR2 photocurrent undergoes a rapid decay to baseline. Close inspection of this portion of the response shows this decay to be biexponential (Fig. 3a) with two distinct time constants (Fig. 3a, bottom insets). This biexponential function rules out any explanation of the ChR2 "off" response being a simple capacitative resistive decay. Rather, there are two separate decay processes with independent time constants.

The dynamics of the ChR2 dark decay for the single light pulses of various durations is analyzed in Fig. 4. When the temporal width of light stimulation pulses is changed (between 1 and 500 ms), there is no discernible change in either decay time constant (Fig. 4a). However, the normalized photocurrent amplitude associated with each decay process does vary

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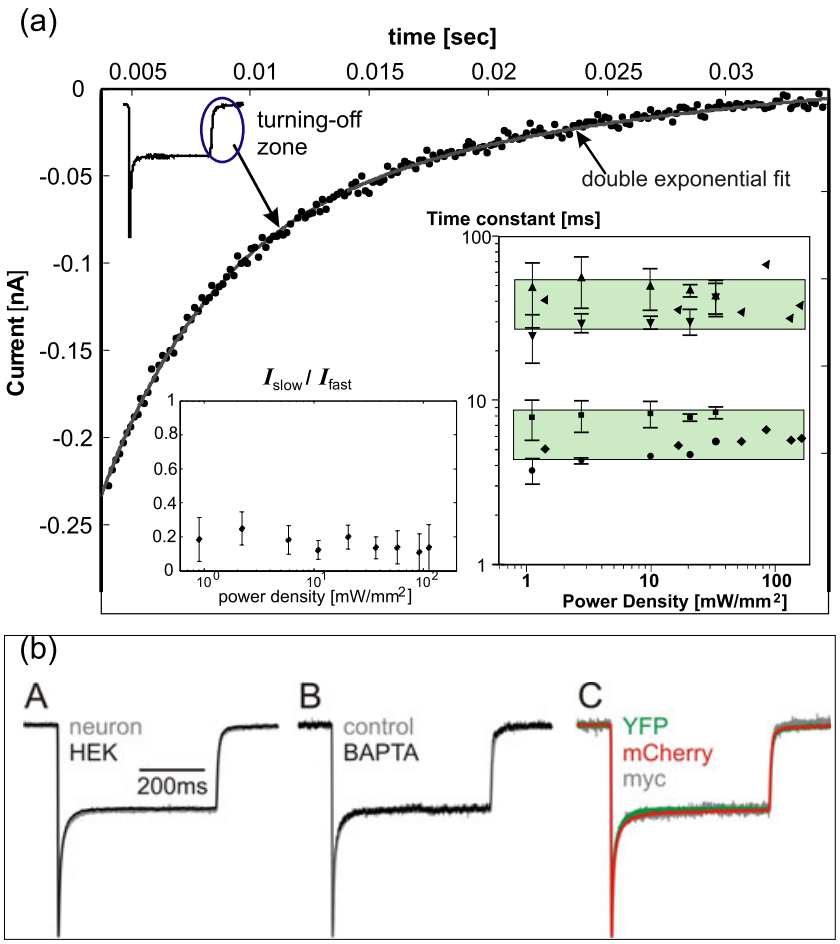


Figure 3. (a) The current decays biexponentially after the light is turned off (ChR2-expressing hippocampal cells). Inset: sets of values for two time constants τ_{fast} and τ_{slow} (bottom right), and the ratio of the amplitudes I_{slow}/I_{fast} (bottom left), for the exponential fit *vs* light intensity, for $n = 4$ cells and three repeated measurements. Bars indicate the standard deviation. Light pulse duration is 500 ms. (b) ChR2 kinetics is intrinsic to the channel: (A) Kinetics of photocurrents recorded in voltage clamp mode (−70mV) in response to a 500 ms light stimulus are identical in neurons and HEK-293 cells. (B) ChR2 kinetics are not altered by the intracellular presence of the fast calcium chelator BAPTA (10 mm). (C) ChR2 kinetics are identical irrespective of whether the channel is tagged by either YFP or mCherry fluorophores, or by a nonfluorescent myc sequence.

with pulse duration (Fig. 4b), indicative of population changes in different ChR2 states as the light stimulus persists. When the turning-off phase of the photocurrent was fitted with a single exponential function, the fit was not as good as in the biexponential case and the turning-off time constant (τ_{off}) was reported in Ishizuka *et al*. (15) to slow down with increasing light duration (*e.g.* $\tau_{off} \approx 5$ ms for 20 ms light pulse, but $\tau_{off} \approx 15$ ms for 1 s pulse). This can now be explained by using results from Fig. 4, where for short light pulses only the fast decay component is present and it has a time constant of about 5 ms. However for longer pulses the slower component amplitude with a time constant 35 ms represents about 20% of the current, so the "composite" time constant which would represent a monoexponential process increases.

Neither time constant of the ChR2 "off" response shows any dependence on light intensity (Fig. 3a, bottom right), and the ratio of the amplitudes of the two exponential functions does not appear to be light intensity dependent (see Fig. 3a, bottom left). This observation is commented on in the Discussion and Appendix 2.

To be certain that the kinetic properties of the ChR2 photocurrent are intrinsic to the channel itself, and not specific to the neuronal context in which it is normally expressed, we also performed patch-clamp recordings on HEK-293 cells transfected with ChR2-YFP. Photocurrents recorded in response to long 500 ms light pulses were qualitatively indistinguishable in HEK-293 cells *vs* hippocampal neurons, and were also quantitatively indistinguishable (see Fig. 3b: A). In both cell types, it is still possible that channel kinetics might be modulated by the internal cellular environment. In particular, changes in internal free calcium levels can directly or indirectly alter the properties of various ion channels (*e.g.* 19,20). To inhibit any such calcium changes, in particular those arising from entry through the open ChR2 channel itself, we loaded neurons with the fast Ca chelator BAPTA (10 mm) *via* the patch pipette. Nevertheless, blocking internal calcium transients in this manner had no effect, either qualitatively or quantitatively, on ChR2 photocurrent kinetics (Fig. 3b: B). Finally, it remains a possibility that the kinetics of the ChR2 channel are affected by the YFP fluorophore attached to its C-terminus. However, replacing this YFP with the mCherry fluorophore had no effect on ChR2 photocurrent kinetics, and we also saw no change in kinetics after replacing the fluorophore sequence with a short nonfluorescing myc tag

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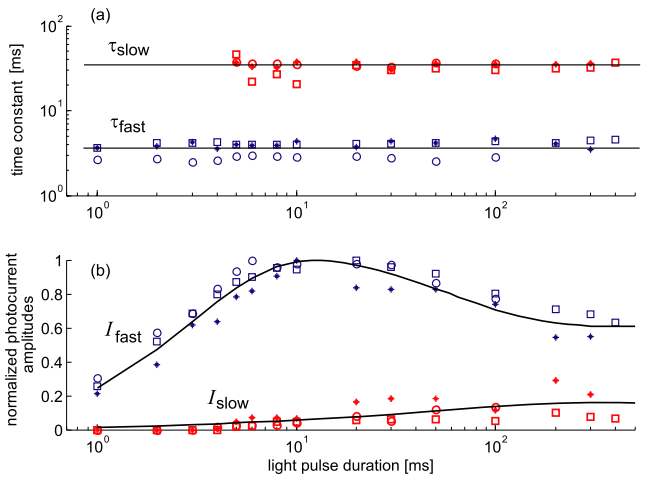


Figure 4. The turning-off phase was fitted with biexponential function with time constants τ_{fast} and τ_{slow} and amplitudes I_{fast} and I_{slow}. The graphs show the dependence of these parameters on the duration of the illumination pulse (1 mW mm^{-2} intensity, $n = 3$ cells). The average values and standard deviations for the time constants are: $\langle\tau_1\rangle = 3.8$ ms, $\sigma_{\tau 1} = 0.65$ ms and $\langle\tau_2\rangle = 34$ ms, $\sigma_{\tau 2} = 5$ ms. The standard deviations calculated for each individual cell are on average only one half of the value calculated for all cells. The lines are calculated using Eqs. (33) and (34) based on the theory given in Appendix 2 (parameters: $I_{max} = 1$ nA, $\varepsilon_1 = 0.5$, $\varepsilon_2 = 0.15$, $\gamma = 0.1$, $G_{d1} = 0.11$ ms^{-1}, $G_{d2} = 0.025$ ms^{-1}, $e_{12} = 0.01$ ms^{-1}, $e_{21} = 0.015$ ms^{-1}, $w_{loss} = 1.1$, $\tau_{ChR} = 0.2$ ms^{-1}).

sequence (these neurons were co-transfected with EYFP) (Fig. 3b: C). The ChR2 photocurrent kinetics described here, and the models we use to fit them, are therefore reflective of intrinsic channel properties of the ChR2 protein.

The three-state model

The three-state photocycle model for ChR2 was first described by Nagel *et al*. (1), and then examined in more detail by Nikolic *et al*. (16). The model, outlined in Fig. 1a, states that upon photon absorption, the molecule which starts in the closed-sensitive state (C) undergoes very fast transition into an excited state (C*). This state results in an open channel state (O), which then spontaneously turns into a closed, but desensitized state (D). Here, the ion channel is closed but the molecule is not ready to photoswitch again. The molecule then goes into a closed but sensitized state (C) after a recovery period. The time constant of the process of opening the ion channel (C $\rightarrow$ C* $\rightarrow$ O) is smaller than 1 ms. The process of closing the ion channel (O $\rightarrow$ D) depends strongly on the extracellular pH_o and takes 10–400 ms. As the opening happens so fast, the model can be reduced to a three-state model (C $\rightarrow$ O $\rightarrow$ D $\rightarrow$ C) (Fig. 1a).

The process of recovery from desensitization (D $\rightarrow$ C) is not yet fully understood. Its duration varies from a few seconds to a minute in the dark (depending on the pH_o), but in the light the process appears to take only 60 ms (6,16). For the three-state model to be correct, the recovery rate must be modulated by light flux. Alternatively, the recovery process could occur only partially *via* the D state, with some molecules going directly from the O to the C state. Having tested both hypotheses, the light flux-modulated recovery seems more plausible.

We have adopted a three-state model as shown in Fig. 1a, which can be presented by a set of two rate equations:

$$\frac{dO}{dt} = \varepsilon FC - G_d O \tag{1}$$

$$\frac{dD}{dt} = G_d O - G_r D \tag{2}$$

where O, D and C are the numbers of ChR2 molecules in a cell which are in the corresponding O, D and C states, $G_d = 1/\tau_d$ and $G_r = 1/\tau_r$ are the rates of the channel closure and recovery of photosensitivity, and $\tau_{d(r)}$ are the corresponding time constants. F is the number of photons absorbed by a ChR2 molecule per unit time and ε is the quantum efficiency. We define the quantum efficiency as the probability that a photon absorbed by the retinal molecule in the ChR2 activates the channel (the exact value depends on the specific distribution of amino acids around the retinal binding pocket and the light wavelength, but a typical value would be $\varepsilon \approx 0.5$ for rhodopsin [21]). If ϕ is the photon flux per unit area, then $F = \sigma_{ret}\phi/w_{loss}$, where w_{loss} is the measure of the loss of the photons from the original flux ϕ resulting from light absorption or scattering by the solution, and σ_{ret} is the retinal cross-section ($\sigma_{ret} \approx 1.2\times10^{-20}$m^2 [14]). If the total number of ChR2 molecules in a cell is N then $N = O + D + C$ and we assume that N stays constant for the observed cell, *i.e.* that there is no ChR2 degeneration. The current measured by whole cell voltage-clamp recording in ChR2-expressing neurons is directly proportional to the number of open channels: $I = Vg_1O$, where g_1 is the conductance of an individual ChR2 ion channel and V is the holding voltage across the cell membrane (here we can assume that the reversal potential for ChR2 is approximately 0 mV [1]).

The three-state model is able to qualitatively reproduce a typical light-evoked response in ChR2-expressing cells, with a rapid increase to the peak current and then a slower decrease to the asymptotic plateau value (see the inset to Fig. 5). Equations (1) and (2) give an analytical solution for the time-dependent currents in the case of constant illumination:

$$I(t) = I_{plat} + ae^{-\lambda_1 t} - be^{-\lambda_2 t}, \tag{3}$$

where:

$$I_{plat} = I_{plat,max}\frac{P}{P+K}, \quad P = \varepsilon F, K = \frac{G_r G_d}{G_r + G_d} \tag{4}$$

$$I_{plat,max} = \frac{G_r}{G_r + G_d} \tag{5}$$

$$\lambda_1 = \alpha - \beta, \quad \lambda_2 = \alpha + \beta \tag{6}$$

$$\alpha = (P + G_r + G_d)/2 \tag{7}$$

$$\beta = \sqrt{((G_r + G_d - P)/2)^2 - G_r G_d} \tag{8}$$

$$a = \frac{P - \lambda_2 I_{plat}}{2\beta}, b = \frac{P - \lambda_1 I_{plat}}{2\beta} \tag{9}$$

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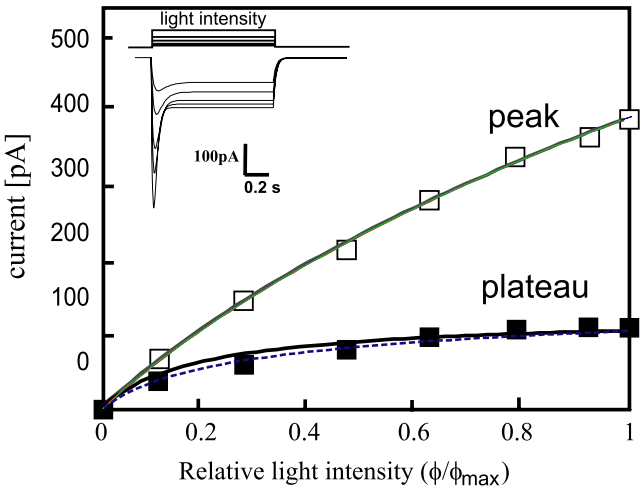


Figure 5. The peak (open squares) and the plateau currents (full squares) in the experiment of Ishizuka *et al.* (15) and our corresponding theoretical values (full lines), according to the three-state model (τ_d = 11 ms and τ_r = 165 ms). The broken line shows the simulation results when G_r depends logarithmically on the light flux, see Eq. (13). Inset: typical response curves (three-state model, Eqs. [3] and [10]) for long light pulses (~1 s) for different light intensities.

When the light is turned off, the current decreases exponentially:

$$I(t) = Ae^{-G_d t}. \quad (10)$$

The major success of the three-state model in the quantitative modeling of the cell's response was in the relatively accurate prediction of the peak and plateau values of the current (shown in Fig. 5). A Michaelis–Menten-type equation for the plateau current given in Eq. (4) was empirically found by Ishizuka *et al.* (15).

The analytical form for the current also gives an opportunity to compare the values of the photocycle parameters obtained from the simulations with those determined from experimentally measured ChR2 currents. For example, in Bi *et al.* (3) the values for λ_1 and λ_2 were determined from curve fitting using the exponential function: $I(t) = a_0 + a_1(1 - \exp(-t/\tau_{act}) + a_2 \exp(-t/\tau_{deact})$, where τs are related to Eq. (6) as $\tau_{act} = 1/\lambda_2$ and $\tau_{deact} = 1/\lambda_1$, and represent the activation and inactivation time constants, respectively. This form we readily recognize in Eq. (3). When λ_1 and λ_2 are determined as fitting parameters from the experimental results, we can estimate the physical parameters $\tau_d = 1/G_d$ and $\tau_r = 1/G_r$ as using equations Eqs. (6)–(8) we get:

$$G_d = (S + \delta)/2, \quad G_r = (S - \delta)/2, \quad (11)$$

$$S = \lambda_1 + \lambda_2 - P, \quad \delta^2 = (\lambda_1 - \lambda_2)^2 - P^2 + 2PS. \quad (12)$$

Precisely determining the values for the ChR2 excitation rate $P(=\varepsilon F)$ might be problematic in a specific experiment. Usually we know the photon flux ϕ, but not F, the actual number of photons absorbed by a molecule per unit time. However, typical experimental light intensities, as in Table 1, give $\lambda_2 \gg \lambda_1$, and $\lambda_2 \gg P_{max}$ (where P_{max} is the maximum activation rate when no photons are lost due to scattering or absorption by the medium), hence the value for P can be estimated and the error will not have a big impact on the final result. Equations (11) and (12) then give: $\tau_d \approx \tau_{act}$ (see Table 1). Results in Table 1 suggest a strong dependence of the two time constants on light intensity. τ_d decreases when ϕ increases, in proportion with $\ln(\phi)$, whereas τ_r first decreases for smaller values for ϕ, but then stabilizes at ~50 ms for higher values (Fig. 6). However, τ_r has much stronger dependence on the intensity of light for lower light intensities: its duration in the dark ($\tau_r \approx 10$ s) is far longer than under 1 mW mm^{-2} illumination ($\tau_r \approx 0.1$ s). Here we have to assume that the recovery process (D $\rightarrow$ C) is light assisted.

Table 1. The open-state (τ_d) and recovery (τ_r) lifetimes according to the three-state model, for the experimental results for the "activation" (τ_{act}) and "deactivation" (τ_{deac}) times from Bi *et al.* ([3], fig. 2G,H). Light pulse duration is 1 s. $P_{max} = \varepsilon\phi\sigma_{ret}$, where ε is estimated to be 0.5 and σ_{ret} is approximately 10^{-8} μm^2.

ϕ ($\times 10^9$ph μm^{-2} s^{-1})	P_{max} (ms^{-1})	λ_1 (ms^{-1})	λ_2 (ms^{-1})	$\tau_{act} = 1/\lambda_2$ (ms)	$\tau_{deact} = 1/\lambda_1$ (ms)	τ_d (ms)	τ_r (ms)
0.22	0.001	0.0018	0.074	13.5	547.7	13.5	576.1
1.8	0.009	0.0095	0.116	8.6	104.8	8.6	114.1
6.2	0.031	0.023	0.204	4.9	42.7	4.9	48.4
13.5	0.068	0.026	0.323	3.1	38.7	3.1	49.8
18.0	0.090	0.029	0.435	2.3	34.8	2.3	47.3

The ChR2 system shows the ability of spontaneous recovery in the dark, because retinal can re-isomerize to the all-*trans* ground state in dark, without the need for special enzymes (5). However the recovery from inactivation is relatively slow in the dark ($\tau_{r,dark} \approx 5$–10 s) (1,5,15). In addition to the re-isomerization, it appears that the process of protonation of one or more amino acids in or near the retinal binding pocket is part of the ChR2 recovery process (1) and hence strong pH_o

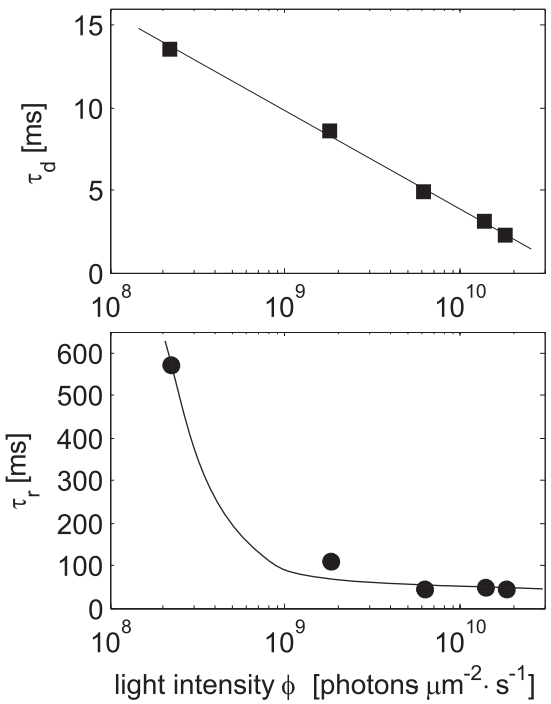


Figure 6. The deactivation (τ_d) and recovery (τ_r) time constants under illumination, *vs* different light intensities (from Table 1).

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dependence of the recovery time (1). Our three-state model can qualitatively reproduce the response to a series of light pulses (see ref. [16], Fig. 5), which is affected by the dark recovery. However, for this the model requires two different values for τ_r: one for light ($\tau_{r,light}$) and one for dark conditions ($\tau_{r,dark}$) meaning that the recovery rate $G_r = 1/\tau_r$ may be dependent upon general light flux. This is not implausible, as in invertebrates chromophore recovery can be assisted *via* absorption of a second photon (22). For example in *Drosophila*, the metarhodopsin is thermally stable and can be re-isomerized back to rhodopsin by absorption of red light (23). After testing several functions we found that a logarithmic dependence for $G_r(\phi)$:

$$G_r(\phi) = G_{r,dark} + c\log(\phi/\phi_0), \tag{13}$$

where c and ϕ_0 are appropriate numerical constants, gives the plateau values in Fig. 5 which are a better match to the experimental values than the case when $G_r(\phi) = \text{const}$.

While the three-state model is able to successfully reproduce several elements of the ChR2 photocurrent, the need to introduce a logarithmic dependence of deactivation on light flux shows that a simple three-state model cannot fully explain the response of ChR2-expressing neurons to a light stimulus. In addition, the three-state model, with its single decay phase, cannot account for the biexponential ChR2 current decay we observed following light offset (Fig. 3), nor can it explain the pH-induced shift in the action spectrum (12). Finally, the parameter space in our three-state model is not big enough to accurately fit the complete ChR2 photocurrent waveform (Fig. 7). To resolve these issues, we have developed a more detailed four-state model of ChR2 kinetics.

The four-state model

The different dark decay time constants seen in our experimental results in Figs. 3 and 4 indicate the potential existence of two open states, O1 and O2. In addition, Ernst *et al.* (12) have pointed to the possible existence of two ChR2 closed states (C1 and C2), and the photocycle experiments of Bamann *et al.* (13) indicate two closed (corresponding to P_0 and P_4) and two open (P_2, P_3) functional channel states. Therefore, if we neglect the short-lived (< 1 ms) intermediate states, a four-state model, shown in Fig. 1b, becomes an attractive proposition.

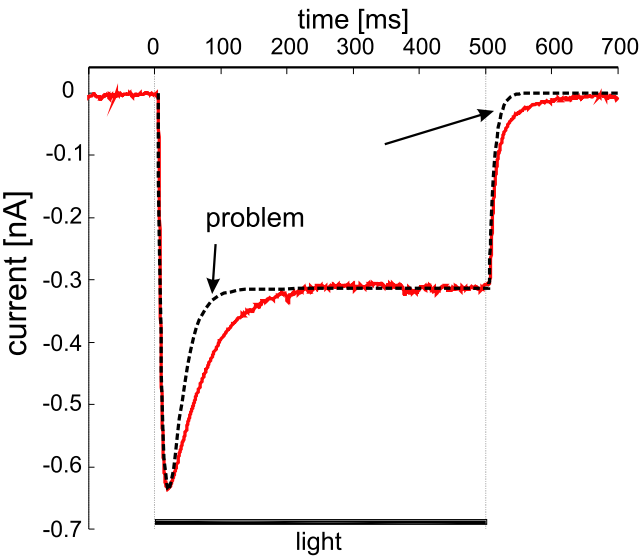


Figure 7. Experimental (full line) and the three-state model simulation results (broken line), with the parameters chosen to fit the peak and plateau of the experiment. The model gives too fast decrease of the currents.

Let us assume that a dark-adapted ChR2 molecule is completely settled in the dark state C1. Upon illumination C1 converts into the open state O1, within 1 ms, going through two fast intermediate states (resolved by the transient spectroscopy and described in Ernst *et al.* [12]). The model allows O1 to convert back to C1. Alternatively, there may be a transition to another open state O2, which could be less conductive than O1. The transition between the two open states is reversible. The O2 state decays to the second closed state C2. C2 thermally converts into C1, but this process is slow (of the order of seconds) in comparison with the other transitions. If in this model we assume that the light-assisted transitions are C1 $\rightarrow$ O1 and C2 $\rightarrow$ O2, then the model can be described by the following set of rate equations:

$$\frac{dO1}{dt} = G_{a1}(t)C1 - (G_{d1} + e_{12})O1 + e_{21}O2 \tag{14}$$

$$\frac{dO2}{dt} = G_{a2}(t)C2 + e_{12}O1 - (G_{d2} + e_{21})O2 \tag{15}$$

$$\frac{dC2}{dt} = G_{d2}O2 - (G_{a2}(t) + G_r)C2 \tag{16}$$

where O1, O2, C1 and C2 are the numbers of the ChR2 molecules in the open states 1 and 2, and closed states 1 and 2, respectively (N = O1 + O2 + C1 + C2), G_{d1} and G_{d2} are the rates of transition O1 $\rightarrow$ C1 and O2 $\rightarrow$ C2, G_r is the rate of thermal conversion of C2 to C1, and e_{12} and e_{21} are the rates of transition between O1 to O2 and vice versa. The photon absorption and isomerization of retinal in rhodopsin is a very fast process ($\sim$200 fs [24]). However, the subsequent conformal change in the protein is a slower process, hence we use the activation rates G_{a1} for the C1 $\rightarrow$ O1, and G_{a2} for the C2 $\rightarrow$ O2 processes. They depend on the quantum efficiencies ε_1 (ε_2) as:

$$G_{ai} = \varepsilon_i F f(t, \tau_{ChR}), \tag{17}$$

where i = 1,2 and τ_{ChR} is the activation time of ChR2 ion channel and the function $f(t,\tau_{ChR})$ is given in Appendix 1. The experimentally measured photocurrent will then be:

$$\begin{aligned} I &= V(g_1O1 + g_2O2) = Vg_1(O1 + \gamma O2) \\ &= I_{max}(o1 + \gamma o2), \end{aligned} \tag{18}$$

where g_1 (g_2) is the conductance of a single ChR2 ion channel in the open state O1 (O2), $\gamma = g_2/g_1$ and V is the holding potential (voltage clamp measurements). $I_{max} = Vg_1N$ is the maximum possible current, when all ChR2 channels are in the O1 state, o1 = O1/N and o2 = O2/N are the fractional occupancies of the open states.

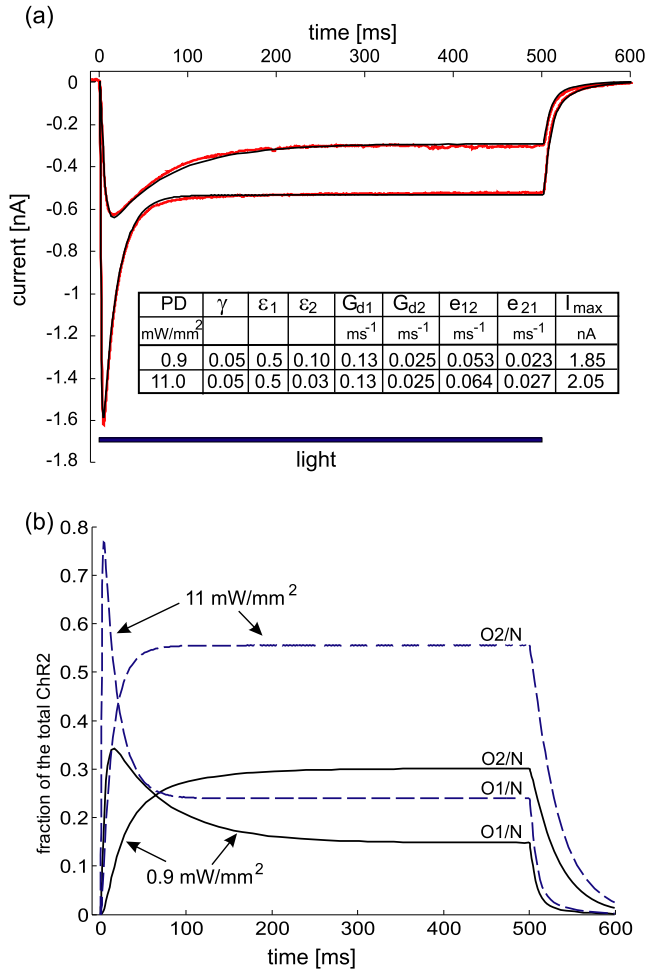
Having developed the four-state model thus far, does it give a better fit to the experimental ChR2 photocurrents than the three-state model? In order to answer this question we have

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performed a numerical fitting procedure using Eqs. (14)–(18), and the results of this modeling are shown in Fig. 8a for two power densities. More details about the relationship between the experimentally derived parameters (*e.g.* the current decay factors Λ_1 and Λ_2) and physical parameters describing the kinetics of ChR2 are given in Appendix 2. The results in Fig. 8a show that the four-state model offers a very good fit to the experimental results, with a reasonable choice of parameters (see the table in Fig. 8).

The results for the distribution of ChR2 molecules between the two open states, both as a function of the duration of illumination and light intensity (Fig. 8b) indicate a light adaptation process. Our fitting parameters show that the ratio of the conductances of the two states is $\gamma \sim 0.1$, hence we could talk about high (O1) and low (O2) conducting states. After the start of the illumination the O1 state is populated first (Fig. 8b). As the illumination continues, the number of ChR2 molecules in that state reaches a peak and then decreases to a steady-state value. At the same time the fraction of the ChR2 molecules in the O2 state is initially low but increases with stimulus time, reaching an asymptotic value. As the light intensity increases, the fraction of molecules in the low conducting state increases faster than the fraction in the high conducting state.

Using the activation rate formulas derived in Appendix 1 Eqs. (24) and (25) and the four-state model Eqs. (14)–(18) we also obtain a good fit for the ChR2 photocurrents recorded in response to short stimulus durations (1–10 ms, Fig. 9). The further rise in current after short light pulses ($t_{light} < 3$ ms) can be explained by the fact that ion channels are not simultaneously opened with photon absorption but the protein conformal change and subsequent channel opening is a slower process. The effect is visible for short pulses as the channels are not closing yet. The introduction of the activation rate function (see Eqs. [24] and [25] in Appendix 2) mathematically describes this process, and gives good results for modeling experimental measurements (Fig. 9). The fitting parameter here for the activation time constant of ChR2 is $\tau_{ChR} = 1.3$ ms, longer than in the case of the Volvox ChR (12), where the average time of the transition from a dark-adapted state to the conducting intermediate in the spectroscopic measurements is about 0.2 ms. The longer activation time in our experiments is probably due to the time constant of our light illumination system (which is about 1 ms), because the same activation rate function (Eqs. [24] and [25]) can describe the light source illumination increase.



PD	γ	ε1	ε2	Gd1	Gd2	e12	e21	Imax
mW/mm^2				ms^{-1}	ms^{-1}	ms^{-1}	ms^{-1}	nA
0.9	0.05	0.5	0.10	0.13	0.025	0.053	0.023	1.85
11.0	0.05	0.5	0.03	0.13	0.025	0.064	0.027	2.05

Figure 8. (a) The experimental photocurrents (red lines) and the fitting results for the four-state model (black lines), for two light intensities: 0.9 and 11 mW mm^{-2} (500 ms light pulse, 470 nm wavelength, −70 mV holding potential). The current decay after the light is turned off with $\Lambda_1 \approx 0.030$ ms and $\Lambda_2 \approx 0.15$ ms ($\Lambda_{1(2)}$ are defined in Appendix 2). The optimized fitting parameters for the two light intensities are given in the table ($G_r = 0.4 s^{-1}$ and $w_{loss} = 1.1$). (b) The time dependency of the occupation levels (o1 and o2) for the O1 and O2 states, for two light intensities.

DISCUSSION

The main modeling results for the proposed four-state model are shown in Fig. 8. The ChR2 molecule may be interpreted as

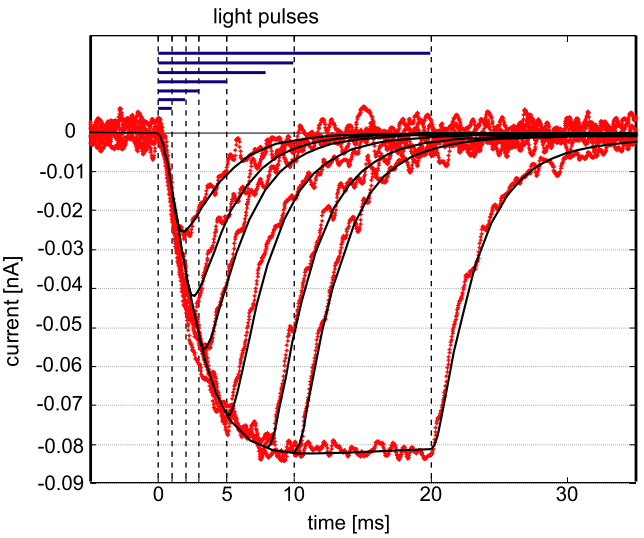


Figure 9. The experimental photocurrents (red crosses) and the model results (full lines) for the four-state model, for short light pulses of 1, 2, 3, 5, 8, 10 and 20 ms. Light intensity is 0.65 mW mm^{-2}, 470 nm wavelength, −70 mV holding potential. The light-off current decay exponents are $\Lambda_1 = 0.0282 \pm 0.0013$ ms^{-1} and $\Lambda_2 = 0.362 \pm 0.0282$ ms^{-1}. The fitting parameters are: $\tau_{ChR} = 1.3$ ms, $\gamma = 0.1$, $\varepsilon_1 = 0.6$, $\varepsilon_2 = 0.14$, $G_{d1} = 0.35$ ms^{-1}, $G_{d2} = 0.02$ ms^{-1}, $e_{12} = 0.01$ ms^{-1}, $e_{21} = 0.02$ ms^{-1}, $w_{loss} = 1.3$, $I_{max} = 0.2$ nA.

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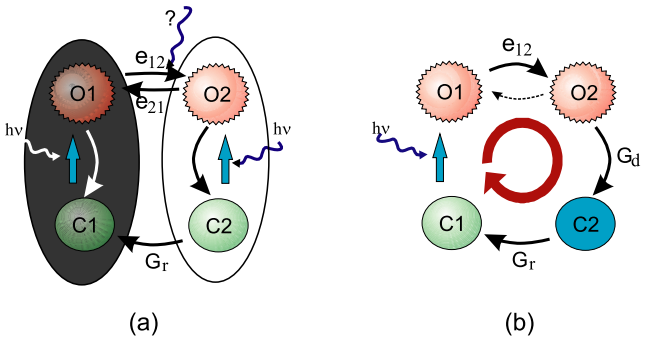


Figure 10. (a) The "branching" model for the ChR2 kinetics. The states could be grouped into the dark-adapted (C1 ↔ O1) and states more preferred by ChR2 into the light-adapted mode (C2 ↔ O2). The transition between the two modes appears to be a combination of light-induced and spontaneous transitions. (b) The "circular" model, where the molecule goes between the states always in one direction.

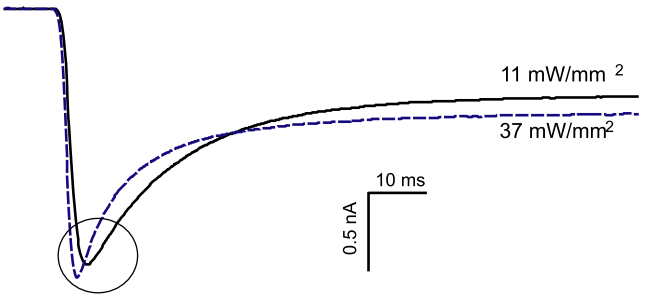


Figure 11. The experimental photocurrents as in Fig. 8, for light intensity 11 and 37 mW mm^{-2}. The crossing between the two curves (circled) indicates stronger deactivation rate in the initial part of the light response for the higher light intensities.

having two sets of intratransitional states: C1 ↔ O1 is more dark-adapted, whereas C2 ↔ O2 is light-adapted, as seen in Fig. 10. ChR2 in the light-adapted state is a mixture of all four states, and ChR2 molecules preferentially accumulate at the O2 state as the duration of illumination increases. A similar behavior has been proposed in the case of the ChR1 photoresponse by Hegemann *et al.* (14). The values obtained for G_{d1}, G_{d2} and γ (see the table in Fig. 8) suggest that the O1 state is a short-lived state with high conductance and the transition C1 → O1 has high quantum efficiency (ε_1). This gives high sensitivity and fast response of the molecules which were previously in the dark for a longer period. In this way the system would maximize the response to a very few photons. If the illumination is prolonged and/or is of high intensity, the system moves to the O2 state, which is long lived and has low conductance. In addition, the transition C2 → O2 (ε_2) has reduced efficiency, hence more molecules are in the closed C2 state.

The communication between the two adaptation modes could be *via* spontaneous transition between O1 and O2, and/or induced by light. Our numerical calculations for e_{12} and e_{21} have shown certain dependence on the light intensity, which could be described with a logarithmic function:

$$e_{12}(\phi) = e_{12\text{dark}} + c_1 \log(1 + \phi/\phi_0), \qquad (19)$$

where $e_{12\text{dark}}$ is the transition rate between O1 and O2 in the dark, c_1 and ϕ_0 are constants (a similar equation is for e_{21}). For example, for our experiments shown in Fig. 8a, the values are $e_{12} = 0.053$ ms^{-1}, $e_{12\text{dark}} = 0.022$ ms^{-1} and $e_{21} = 0.023$ ms^{-1}, $e_{21\text{dark}} = 0.011$ ms^{-1} for 0.9 mW mm^{-2} light intensity. This model suggests a larger deactivation rate for higher light intensities, and that is exactly what was seen in our experiments (see Fig. 11). The circled region shows the part of the photocurrent where the rate of decrease for higher power density is clearly larger causing the two curves to cross. A similar behavior can be seen in other reported experimental results (*e.g.* fig. 2C in Bi *et al.* [3]). Nevertheless, more experimental evidence is needed to establish whether the transition occurs after the retinal (in the open states O1 or O2) absorbs a photon, and whether the subsequent reisomerization changes the state of the protein.

The spectroscopy measurements of Bamann *et al.* (13) identified four intermediates in the ChR2 photocycle (P_1, P_2, P_3 and P_4). The first one is a very short-lived intermediate (< 1 ms) and functionally is perhaps less significant. P_2 and P_3, with absorption maximum at 520 nm are identified as the cation conducting states of the channel. They could be interpreted as O1 and O2 states in the four-state model. Finally, the relatively long-lived P_4 state could be the C2 state. The numerical calculations and analytical relationships given in Appendix 2 suggest that the decay rate G_{d1} is significantly faster than the decay rate O2 → C2, G_{d2} (see Fig. 8a table with the fit parameters). The conductance of the O2 state seems to be only a fraction of the O1 state.

We can also classify the proposed models for the ChR2 photocycle into two groups: (1) branching, as can be seen in Fig. 10a and (2) circular, as can be seen in Fig. 10b, photocycles. The branching models allow for different paths to be taken from a given state, whereas the circular models have a unidirectional flow through the available states. For example, in the branching model, a molecule in state O1 can take one of two branches: a transition back to C1 or a transition to O2. A problem with the branching model is that C1 might not be directly accessible from O1 due to the possible need for protonation of one or more amino acids in ChR2 apoprotein from the extracellular side. The problem with the circular models is that they require for the recovery time constant to be light dependent, *i.e.* that the process in some way involves secondary photon absorption.

The adopted four-state model can describe the impact of the light duration on the off-kinetics as shown in Fig. 4. The mathematical description of the off-kinetics is given in Appendix 2. The light-off photocurrent decays biexponentially: $I_{\text{off}} = I_{\text{slow}} \exp(-\Lambda_1 t) + I_{\text{fast}} \exp(-\Lambda_2 t)$, see Eq. (33), where the exponents are given by Eq. (28). The amplitudes for the slow and fast component were determined from the experimental recordings for different light pulse durations (1–500 ms) and normalized values are shown in Fig. 4b. The amplitude of the slow component is very small for short pulses (< 5 ms) almost completely under the noise floor, hence the decay current is virtually monoexponential. For longer pulses the slow component increases and saturates, whereas the fast component goes through a maximum value, then decreases and reaches an asymptotic value, where the ratio $I_{\text{slow}}/I_{\text{fast}} \sim 0.20$. The lines in Fig. 4b are the theoretical

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calculations based on Eq. (36). The ratio I_{slow}/I_{fast} depends on the light pulse duration, until it reaches asymptotic values. However, when the two open states are in equilibrium, *i.e*. for light pulses longer than about 200 ms, the ratio does not show significant dependence on the light intensity, as shown in the experimental results (Fig. 3) and explained theoretically (Eq. [36]).

The retinal molecule has planar structure in its isolated form according to the crystallographic measurements, but in the rhodopsin protein environment it can be twisted due to the steric interaction at the binding pocket (25). There is a specific structure of retinal which depends on the exact arrangement of the amino acids in and around the binding pocket. The chromophore is able to undertake ground-state isomerization around the $C_{13} = C_{14}$ and $C_{15} = N_{16}$ double bonds in the last step of the photocycle and dark-adaptation process for bacteriorhodopsin (25). In the case of ChR2 it seems that the dark adaptation process has a time constant which is on the order of $\tau_{r,dark} \approx 2 - 10$ s for $pH_o = 7.4$. This mechanism would provide a transition route between the light- and dark-adapted modes *via* closed states (C2 $\rightarrow$ C1). The process is relatively slow in comparison with other time constants in the system. The recovery time depends on the extracellular pH: it is slower for higher pH_o and faster for lower (1). It is interesting that dependence is opposite in ChR1—the peak current recovery is faster for higher pH_o (14). The observed pH_o dependency of the ChR2 activation could be due to the effect of the extracellular pH on the stability of the protonated Schiff base and the release of a proton into the extracellular part of the channel. The involvement of water molecules in this process was suggested in Tajkhorshid *et al*. (25).

CONCLUSION

The three-state and four-state models are the two simplest models which could explain the kinetics of the ChR2-expressing cells in the voltage clamp measurements. The three-state model is successful in predicting the peak and steady-state current and for analyzing fast kinetics of ChR2. It provides a simple analytical form for the photocurrents which allows for the calculation of the two physical properties of the system, the deactivation and recovery time constants. When combined with the experimental results, the three-state model gives a prediction of the ChR2 deactivation and recovery rates dependent on the light intensity. However, the model is not accurate enough for quantitative calculations and cannot explain the biexponential decay of the light-off current. The four-state model is represented with two closed (C1 and C2) and two open states (O1 and O2). It shows good results for quantitatively reproducing the experimental photocurrents. The results describing the distribution of the different states in the ChR2 population, as a function of light pulse duration and intensity, suggest a built-in adaptation mechanism. This mechanism represents an interesting example of adaptation, which is established at the individual molecular biology level. The light adaptation function is achieved in higher organisms in a much more complex fashion using calcium feedbacks or rhodopsin bleaching processes.

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APPENDIX 1

The activation rate of the ChR2 ion channels can be obtained in the following way. If the number of photons absorbed by the retinal in a ChR2 molecule per unit time is F and the quantum efficiency of the ion channel opening is ε, then the rate of generating an isomerized retinal which is subsequently causing ion channel opening is εF. The probability for creation of such a retinal molecule during the period dt is $\varepsilon F dt$. Assume now that the activation of the ChR2 protein molecule is a random process with a probability per unit time of $g_a = 1/\tau_{ChR}$. If N_0 retinals are simultaneously isomerized at the instant $t = 0$, there will be $N(t)$ still inactive protein molecules after time t, and that number will change in the next dt for: $dN = -Ng_a dt$. By integrating this equation one gets: $N(t) = N_0 \exp(-t/\tau_{ChR})$. Now the probability that the protein molecule is activated during the time t after the photon was absorbed by the retinal in its core is:

$$p_{active}(t) = 1 - \exp(-t/\tau_{ChR}). \tag{20}$$

If the light illumination occurs from $t = 0$ to $t = t_{light}$, then the number of activated molecules at the time $t < t_{light}$ (assuming that they do not deactivate) is given by the convolution of εF and $p_{active}(t)$ functions:

$$\begin{aligned} X_{active}(t) &= \int_0^t \varepsilon F(1 - \exp(-(t-\tau)/\tau_{ChR}))d\tau \\ &= \varepsilon F[t - \tau_{ChR}(1 - \exp(-t/\tau_{ChR})]. \end{aligned} \tag{21}$$

Similarly, the number of active molecules at the moment t, where $t > t_{light}$ is:

$$X_{active}(t) = \int_0^{t_{light}} \varepsilon F(1 - \exp(-(t-\tau)/\tau_{ChR}))d\tau \tag{22}$$

$$= \varepsilon F[t_{light} - k\exp(-t/\tau_{ChR})] \tag{23}$$

where $k = \tau_{ChR}(\exp(t_{light}/\tau_{ChR}) - 1)$. From Eq. (23) we see that the total number of activated molecules after the light pulse of duration t_{light} becomes $\varepsilon F t_{light}$, as expected, but the value is exponentially approached with the time constant τ_{ChR}. Finally the rate of activation $G_a(t) = dX_{active}/dt$ will be:

$$G_a(t) = \varepsilon F[1 - \exp(-t/\tau_{ChR})], \quad \text{for } t < t_{light} \tag{24}$$

$$= \varepsilon F[\exp((t - t_{light})/\tau_{ChR}) - \exp(-t/\tau_{ChR})], \quad \text{for } t > t_{light}. \tag{25}$$

APPENDIX 2

In the case of the four-state model Eqs. (14) and (15), when the light is turned off, reduce to:

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$$\frac{\mathrm{dO1_{off}}}{\mathrm{d}t} = -(G_{\mathrm{d1}} + e_{\mathrm{12dark}})\mathrm{O1_{off}} + e_{\mathrm{21dark}}\mathrm{O2_{off}}$$
$$\frac{\mathrm{dO2_{off}}}{\mathrm{d}t} = e_{\mathrm{12dark}}\mathrm{O1_{off}} - (G_{\mathrm{d2}} + e_{\mathrm{21dark}})\mathrm{O2_{off}}. \quad (26)$$

The solutions are:

$$\mathrm{O1_{off}} = A_{11}\exp(-\Lambda_1 t) + A_{12}\exp(-\Lambda_2 t)$$
$$\mathrm{O2_{off}} = A_{21}\exp(-\Lambda_1 t) + A_{22}\exp(-\Lambda_2 t) \quad (27)$$

where:

$$\Lambda_{1,2} = b \mp c, \quad (28)$$

$$b = \frac{(G_{\mathrm{d1}} + G_{\mathrm{d2}} + e_{\mathrm{12dark}} + e_{\mathrm{21dark}})}{2}$$

and

$$c = \sqrt{b^2 - (G_{\mathrm{d1}}G_{\mathrm{d2}} + G_{\mathrm{d1}}e_{\mathrm{21dark}} + G_{\mathrm{d2}}e_{\mathrm{12dark}})}.$$

Two useful relationships follow from the previous equations:

$$\Lambda_1 + \Lambda_2 = G_{\mathrm{d1}} + G_{\mathrm{d2}} + e_{\mathrm{12dark}} + e_{\mathrm{21dark}} \quad (29)$$

$$\Lambda_1\Lambda_2 = G_{\mathrm{d1}}G_{\mathrm{d2}} + G_{\mathrm{d1}}e_{\mathrm{21dark}} + G_{\mathrm{d2}}e_{\mathrm{12dark}}. \quad (30)$$

The current after the light is turned off is:

$$I_{\mathrm{off}} = V(g_1 O1 + g_2 O2) = Vg_1(O1 + \gamma O2) \quad (31)$$

$$= Vg_1[(A_{11} + \gamma A_{21})\exp(-\Lambda_1 t) + (A_{12} + \gamma A_{22})\exp(-\Lambda_2 t)]$$

$$= I_{\mathrm{slow}}\exp(-\Lambda_1 t) + I_{\mathrm{fast}}\exp(-\Lambda_2 t) \quad (32)$$

where $I_{\mathrm{slow}} = Vg_1A_{\mathrm{slow}}$ and $I_{\mathrm{fast}} = Vg_1A_{\mathrm{fast}}$ are the amplitudes for the slow (Λ_1) and fast (Λ_2) components of the off-current. From (26), (27) and (32) we get:

$$A_{\mathrm{slow}} = A_{11} + \gamma A_{21} = \frac{[(\Lambda_2 - (G_{\mathrm{d1}} + (1-\gamma)e_{\mathrm{12dark}})]O1_0 + [(1-\gamma)e_{\mathrm{21dark}} + \gamma(\Lambda_2 - G_{\mathrm{d2}})]O2_0}{\Lambda_2 - \Lambda_1} \quad (33)$$

$$A_{\mathrm{fast}} = A_{12} + \gamma A_{22} = \frac{[(G_{\mathrm{d1}} + (1-\gamma)e_{\mathrm{12dark}} - \Lambda_1]O1_0 + [\gamma(G_{\mathrm{d2}} - \Lambda_1) - (1-\gamma)e_{\mathrm{21dark}}]O2_0}{\Lambda_2 - \Lambda_1} \quad (34)$$

where $O1_0$ and $O2_0$ are the numbers of the ChR2 molecules in the open states O1 and O2, respectively, at the moment when the light is turned off. The experimental results for the fast decaying and slow decaying components of the light-off current are shown in Fig. 4b, which shows that for the very short pulses (a few milliseconds) $A_{\mathrm{fast}} \gg A_{\mathrm{slow}}$. For very short pulses: $O2_0 \approx 0$, see Fig. 8b, hence from (34) we get a condition:

$$\Lambda_2 \approx G_{\mathrm{d1}} + (1-\gamma)e_{\mathrm{12dark}}. \quad (35)$$

Equations (29), (30) and (35) were used in initial assessment of the relevant parameters in the fitting process.

During light illumination the equilibrium is reached after about 100–200 ms and then we have $e_{12}O1 \approx e_{21}O2$, as the number of transitions $O1 \rightarrow O2$ (Fig. 1b) has to be equal to the number of transitions $O2 \rightarrow O1$, because the value for the recovery rate G_r is much smaller than other transition rates. Therefore the ratio of the amplitudes of the slow and fast components of the light-off current from Eqs. (33) and (34) becomes:

$$\frac{A_{\mathrm{slow}}}{A_{\mathrm{fast}}} = \frac{\Lambda_2 - G_{d1} - (1-\gamma)e_{\mathrm{12dark}} + [\gamma(\Lambda_2 - G_{d2}) + (1-\gamma)e_{\mathrm{21dark}}]\mathrm{O2_0/O1_0}}{G_{d1} + (1-\gamma)e_{\mathrm{12dark}} - \Lambda_1 + [\gamma(G_{d2} - \Lambda_1) - (1-\gamma)e_{\mathrm{21dark}}]\mathrm{O2_0/O1_0}} \quad (36)$$

This ratio is almost independent of the power density of light, because all Λ and G_d parameters must have light-independent values in the dark, and using Eq. (19), the ratio:

$$\frac{O1_0}{O2_0} = \frac{e_{\mathrm{21dark}} + c_2\ln[1 + (\phi/\phi_0)]}{e_{\mathrm{12dark}} + c_1\ln[1 + (\phi/\phi_0)]}$$

is only weakly light dependent, as found in the simulations, and also this ratio is multiplied by a small factor.

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